Counted but not known

Machine-only building coverage in rural, tribal and wildfire-exposed tracts; two measurement artifacts in the transport metric; and a reproducible methodology — Bias Bounty: Mapping Equity Challenge

Zindi username **dodo25** · associated submission **pwM1ZRTe** (public 0.06391, private 0.06457) · code and data: github.com/tdogukan470-jpg/zindi-bias-bounty-dodo25 · 1 November 2026, version 3 (supersedes the 29 October and earlier 1 November versions)

Executive summary

This document is submitted for both special prizes. **Part I** is the Bias Discovery; **Part II** documents the leaderboard methodology. Every number in both parts is computed from the challenge bundle and from our own scored submissions; no TIGER, Microsoft, HIFLD or CBP data was used anywhere, for scoring or for the discovery analysis. Every table and figure is regenerated by the scripts listed in Section 12, and the final submission is rebuilt by one of them.

Finding 1 — Counted but not known. Across the four scored regions (9,386 tracts in the bundle, 38.8 million residents), the Overture building layer in rural, tribal and wildfire-exposed tracts is *complete by count and empty by content*. In the average rural tract **73%** of buildings are Microsoft machine-detected footprints with no type, no class and almost never (2%) a second source confirming them; in the average urban tract the figure is 30% (building-weighted: 71% of all rural buildings, 32% of all urban ones). In tribal statistical areas the share is **80%**, in the top wildfire-exposure decile **70%**. Only **4%** of buildings in the average rural tract carry any attribute, against 30% in the average urban tract. In the 1,816 tracts where more than 90% of buildings are machine-only, the footprint count agrees with a second footprint inventory (the USFS building count in the strata table) to a median ratio of 1.01 — the same as everywhere else — while the median attribute share is 0.6%. **7.0 million people** live in those tracts (40% of them in rural tracts, 20% in tribal areas, 22% in the top wildfire decile). We tested the pattern against the hidden reference through the leaderboard: the organiser’s composite gap is itself *higher* in machine-only tracts (a probe that raised predicted gaps there improved RMSE from 0.0790 to 0.0765; its mirror image worsened it to 0.0806). So the scorecard captures the pattern **in level but not in kind**: it cannot tell a tract whose buildings are known from one whose buildings are only counted, and its count-based building component can be satisfied by ingesting more machine output without anyone ever looking at the place. The effect sizes (Cohen’s d, pooled, 0.97–1.27) are as large as the rural–urban disparity in the score itself (d = 1.15 on our final submission). For evacuation planning, shelter identification and damage assessment, a footprint without a type is not coverage.

Finding 2 — A boundary double-count inflates rural transport gaps. Tract boundaries follow highways. A reference road on a shared boundary is counted in both tracts; Overture’s slightly offset copy falls in one, so the neighbour shows a transport gap that does not exist. Under our reconstruction the artifact averages **+0.066** on the transport component and is **1.81× larger in rural tracts** — not because rural tracts have more boundary roads (they have fewer: 34% of their reference length lies on a boundary, against 58% in urban tracts) but because urban tracts sit far inside the clipped-at-zero region of the gap and rural tracts sit at the edge of it. It explains about **29% of the rural–urban difference** in our final submission’s mean coverage

gap. One leaderboard probe supports partial double counting in the organiser's pipeline; it does not establish the magnitude.

Finding 3 — Ramps. Overture carries interchange ramps as `subclass = link` within the four counted road classes; TIGER codes ramps S1630, outside the reference. Counting ramps hides small gaps, 3.7× more in rural tracts and 0.2× in tribal areas. This finding is documented in the data but was not tested on the leaderboard.

Methodology. With no training labels, we reconstructed the transport component deterministically from Overture geometry, used interpretable proxies for the building and POI components, and calibrated with a leaderboard-probe calculus whose predictions matched realised public scores to within 0.00015 on the last four re-fits. The public score went from 0.0790 to **0.0639** (private 0.0646). The final submission is rebuilt from the shipped scored vectors by `code/make_t24.py` to within $2 \cdot 10^{-5}$. We also report where the calculus overfit the public split and a fact we learned only after the close: Eastern Washington is excluded from the public board but present in the private one. One caveat: the leaderboard entry for dodo25 shows an early file because of a final-selection error on our side; the method's actual scores are those of `pwM1ZRTe`, verifiable on the submissions page.

Definitions used throughout

- **Bundle**: the challenge data as distributed (`<region>-tracts`, `-strata`, `-acs`, `-bldg-tract`, `-pois-tract`, Overture roads).
- **Reference proxy** (ours, Part II): Overture road segments carrying a route tag for an Interstate, US or State route (`core`) or a Texas FM/RM route (`fm`), excluding `subclass = link`. **Numerator**: Overture `motorway`, `trunk`, `primary`, `secondary`, all subclasses. **L0 / L8 / L25**: length inside the tract polygon / inside the polygon buffered by 8 m / by 25 m. **Lbd8**: length within 8 m of the tract boundary.
- **Transport defined** (model, Part II): core-route reference within the buffers > 100 m (6,269 scored tracts). **Transport defined (artifact analysis, Part I §2–3)**: core + FM reference, strictly inside, > 100 m (6,707 tracts). Both are stated where used.
- **TIGER codes**: S1100 primary road, S1200 secondary road, S1630 ramp (Census MAF/TIGER Feature Class Codes).
- **ML-only share, attribute share, multi-source share**: Section 1.2. **OTSA**: Oklahoma Tribal Statistical Area (`tribal_statistical` in the strata table).
- **Welch t**: two-sample t-test without equal-variance assumption. **Cohen's d**: $(\text{mean}_1 - \text{mean}_2)$ / pooled SD, pooled SD $= \sqrt{((n_1-1)s_1^2 + (n_2-1)s_2^2)/(n_1+n_2-2)})$. $|t| > 5$ implies $p < 10^{-6}$.
- **Tract mean** vs **building-weighted**: a share averaged over tracts vs computed over all buildings; stated where it matters.
- **Probe calculus, Gram matrix, leave-one-out**: Section 9.

PART I — BIAS DISCOVERY

1. Finding 1: Counted but not known

1.1 What the scorecard measures, and what it cannot

The challenge's building component compares the number of Overture building footprints in a tract with a reference count (Microsoft US Building Footprints, per the challenge data page). The automated scorecard then reports disparity ratios of the composite score across fixed strata (rural/urban, tribal, SVI, CVI, hazard). Two things are invisible to this design:

1. **Content.** A count does not know whether a footprint is a house, a school, a fire station, a barn or a carport. Overture carries this as `subtype`/`class`. Machine-detected footprints carry none of it.
2. **Provenance.** Overture's building theme merges OpenStreetMap, Esri Community Maps and machine-detected roofprints from Microsoft and Google; its documentation gives OpenStreetMap the highest conflation priority and uses the machine sources "to fill in coverage" (Overture buildings guide, retrieved 1 Nov 2026). In the bundle, 2% of Microsoft footprints are matched by a second source, against 72% of OpenStreetMap buildings. The share of Microsoft footprints in a tract is therefore a close measure of how much of its building stock no OpenStreetMap contributor has mapped (Esri and Google together are 6% of the layer). A count treats such a footprint and a surveyed, typed, community-verified building as the same unit.

The scorecard can therefore say *that* a tract is under-covered; it cannot say that the buildings it does have are unknown. Section 1.4 tests the first half of that sentence against the hidden reference.

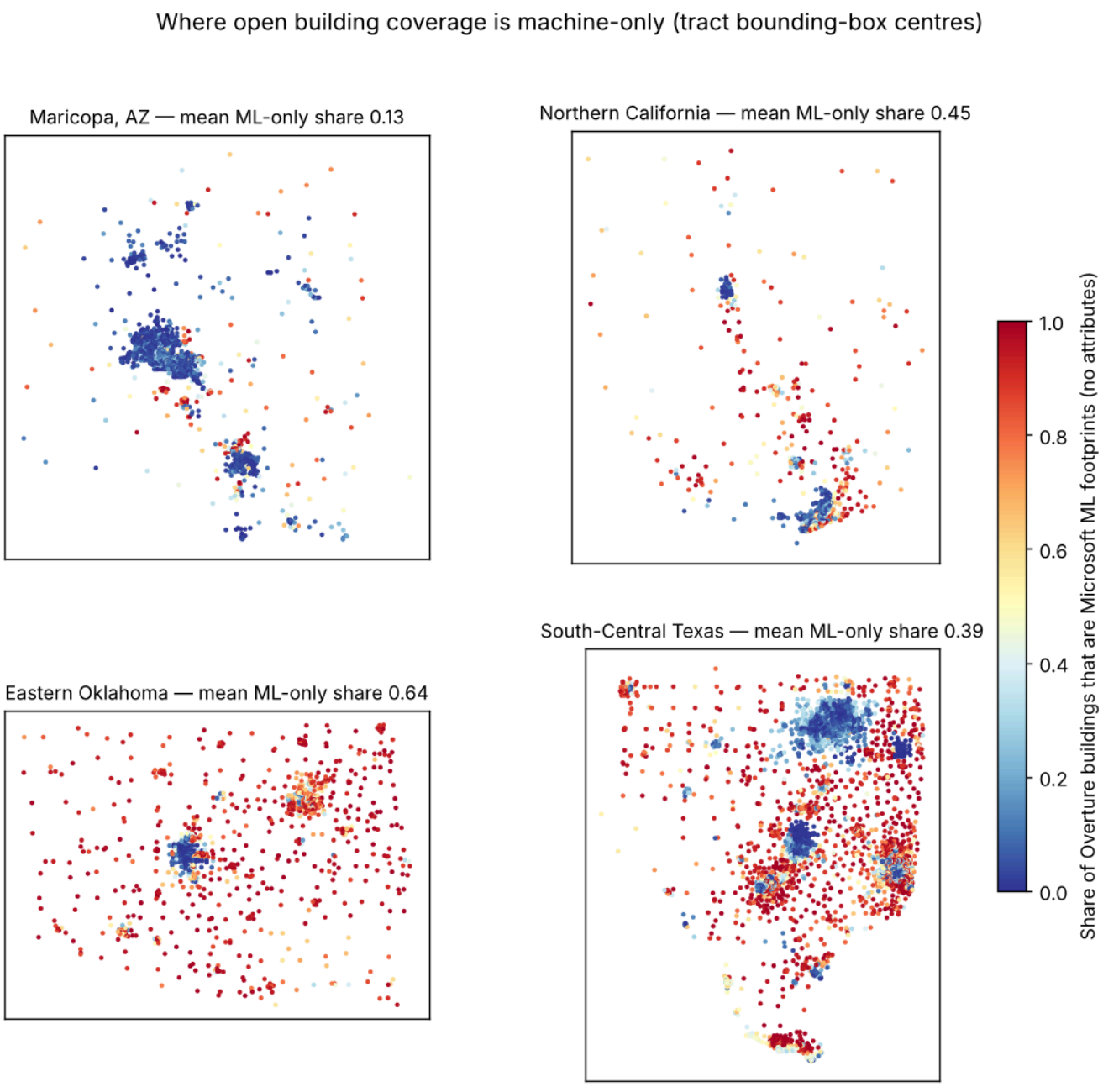
1.2 Measures and data

All measures come from the bundle's per-tract building aggregates (`src`, `subtype`, `class`, `n`, `n_multisrc`), the POI table (`cat`, `confidence`, `src`), the tract table (`pop_total`, `ur_class`) and the strata table (`tribal_statistical`, `usfs_Exposure_mean`, `usgs_wildfire_ever`, `svi_overall`, `svi_minority`, `cvi_overall`, `epht_heat_days_summer`, `usfs_BuildingCount_sum`). For each tract:

- **ML-only share** = buildings with `src = "Microsoft ML Buildings"` / all buildings. 100% of these footprints have a null `subtype`.
- **Attribute share** = buildings with a non-null `subtype` / all buildings. OpenStreetMap buildings carry a `subtype` 40% of the time (the rest are plain `building=yes` outlines), so this share is bounded by how much of a tract is community-mapped; the next measure does not depend on tagging.
- **Multi-source share** = `n_multisrc` / `n`: buildings matched by more than one source.
- **Count coverage** = Overture buildings / `usfs_BuildingCount_sum`. The USFS Wildfire Risk to Communities building count is a 30 m raster count of footprints; its second edition derives it from the USA Structures and ONEGEO footprint inventories (metadata, retrieved 1 Nov 2026), its first from Microsoft footprints. The bundle does not state which edition it carries, so we use it only as a second footprint inventory, never as ground truth.
- POI measures: share of POIs whose `src` is `meta` (Facebook business pages), mean `confidence`, share with `confidence < 0.5`, and presence of a fire-department POI.

Groups are the scorecard's own strata: Rural (`ur_class`), tribal statistical area (`tribal_statistical > 0`), wildfire exposure top decile (`usfs_Exposure_mean`), ever burned (`usgs_wildfire_ever`), SVI ≥ 0.75, CVI top decile, summer heat days top decile. All measures are tract means unless marked building-weighted. The full table is `results/quality_gap_table.csv` (42 rows: six measures by seven strata), the region table `results/quality_gap_regions.csv`, and every other number in this section is in `results/quality_gap_numbers.json`.

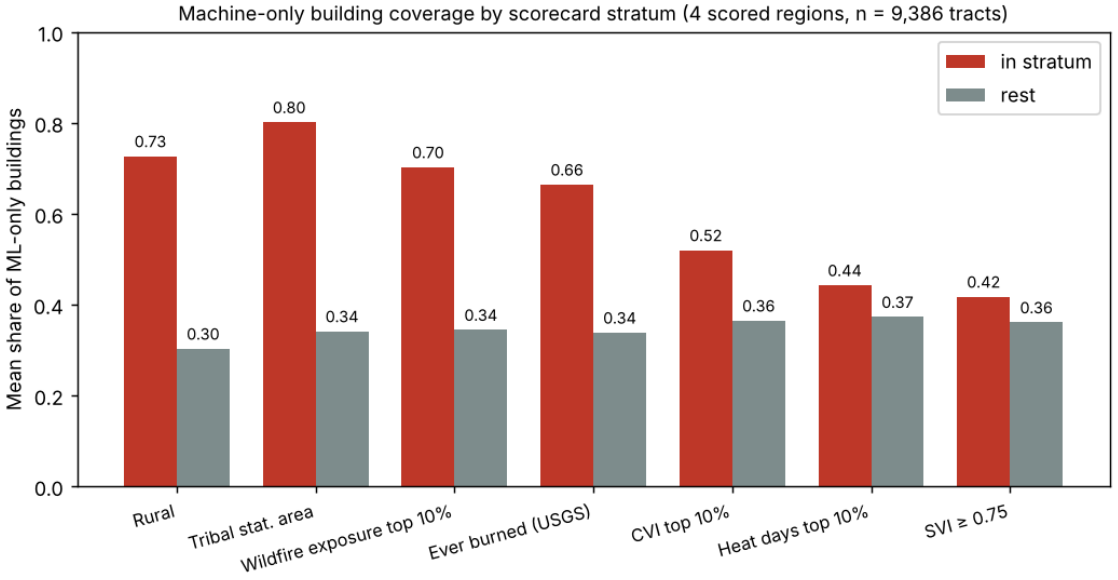
1.3 Results



Share of Overture buildings that are machine-only footprints, by tract (bounding-box centres). Blue = community-mapped, red = machine-only. Eastern Oklahoma, most of which lies in tribal statistical areas, is almost entirely red.

Table 1. Building-layer content by scorecard stratum (four scored regions, n = 9,386 tracts; tract means; d = Cohen's d with pooled SD; all |t| > 5 except where p is shown).

Stratum	n	ML-only share, in stratum	rest	ratio	d	Attribute share, in	rest	d	Multi-source share, in	rest	d
Rural	1,712	**0.727**	0.303	2.40	**1.22**	**0.041**	0.296	−0.80	**0.141**	0.529	−1.23
Tribal statistical area	794	**0.801**	0.341	2.35	**1.27**	0.067	0.267	−0.61	0.124	0.489	−1.09
Wildfire exposure top 10%	939	**0.702**	0.344	2.04	**0.97**	0.047	0.272	−0.69	0.154	0.492	−1.01
Ever burned (USGS)	1,211	0.664	0.338	1.96	0.89	0.072	0.276	−0.63	0.208	0.495	−0.85
CVI top 10%	938	0.520	0.365	1.43	0.41	0.069	0.270	−0.61	0.378	0.467	−0.26
Heat days top 10%	997	0.443	0.373	1.19	0.18	0.097	0.268	−0.52	0.515	0.451	+0.18
SVI ≥ 0.75	3,233	0.417	0.361	1.16	0.15	0.210	0.270	−0.18	0.440	0.468	−0.08 ($p = 2{\cdot}10^{-4}$)



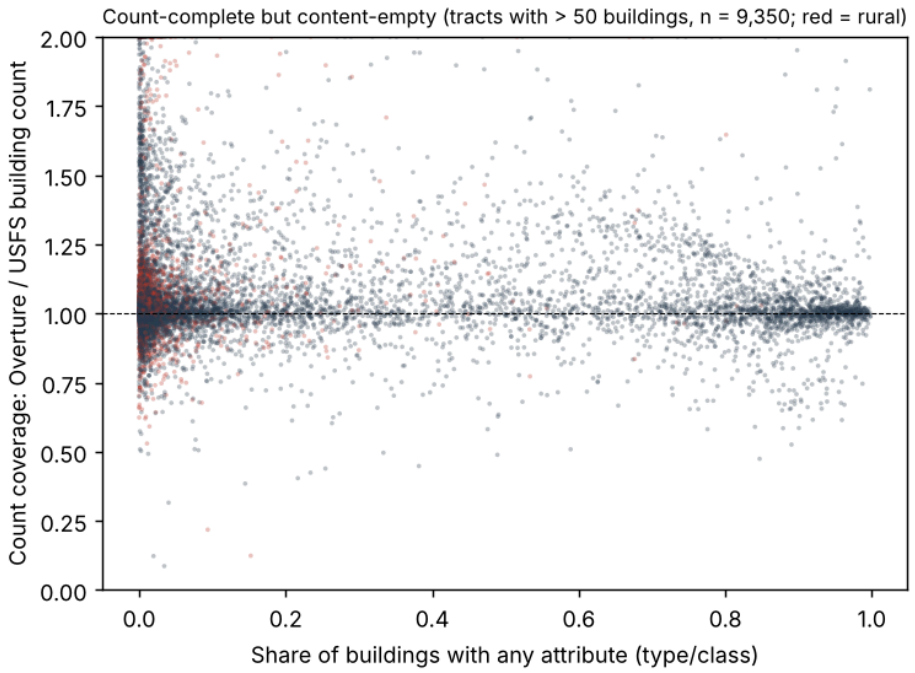
Machine-only share by stratum. The rural, tribal and wildfire differences (0.36–0.46) are six to eight times the socio-economic (SVI) difference (0.06).

Table 2. The same measures by region and stratum (tract means), with resident population.

Region	Stratum	Tracts	ML-only	Attribute	Multi-source	Population
Eastern Oklahoma	Tribal statistical area	792	0.80	0.067	0.12	2,627,624
Eastern Oklahoma	Rural	411	0.87	0.025	0.06	1,326,111
Eastern Oklahoma	Wildfire top 10%	161	0.90	0.022	0.05	528,047
South-Central Texas	Rural	991	0.73	0.035	0.14	3,622,198
South-Central Texas	Wildfire top 10%	527	0.75	0.026	0.13	1,940,153
Northern California	Rural	147	0.73	0.047	0.16	497,511
Northern California	Wildfire top 10%	97	0.75	0.045	0.16	340,174
Maricopa, AZ	Rural	163	0.37	0.115	0.32	536,056
Maricopa, AZ	Wildfire top 10%	154	0.30	0.145	0.36	525,327
All four regions	*Rural AND wildfire top 10%*	836	0.73	0.037	0.14	2,910,304
All four regions	*Tribal AND rural*	366	0.87	0.026	0.06	1,189,189

Regional means of the ML-only share: Eastern Oklahoma 0.64, Northern California 0.45, South-Central Texas 0.39, Maricopa 0.13. The pattern recurs inside every region and is strongest where strata overlap.

Count-complete, content-empty. In the 1,816 tracts where more than 90% of buildings are machine-only, the median count coverage against the USFS inventory is **1.009**; in all other tracts it is 1.020. A count sees no difference. In the same tracts the median attribute share is **0.006**. The scatter below shows the two axes against each other (tracts with more than 50 buildings).



Count coverage against attribute share, tracts with more than 50 buildings (n = 9,350; red = rural). Rural tracts sit on the left: complete by count, empty by content.

No community mapping at all. In 1,414 tracts (15% of tracts, 5.37 million residents) fewer than 5% of buildings come from OpenStreetMap. The share of such tracts is 36% among rural tracts versus 10% elsewhere, 37% versus 13% for tribal areas, and 34% versus 13% for the wildfire decile.

In absolute terms (building-weighted). Rural tracts contain 5.38 million Overture buildings; **5.16 million (95.9%)** have no type or class. Non-rural tracts contain 12.7 million, of which 73% have none.

1.4 What the hidden reference says: captured in level, not in kind

A natural objection: Microsoft's footprint product is the organiser's building reference and the source of these footprints, so the count-based gap might be near zero exactly here, and the scorecard would report *good* coverage where content is worst. We tested this on the leaderboard rather than assume it. From the transport-only reconstruction `t3` (public 0.07904, private 0.07911) we scored the same perturbation in both directions. The perturbation is linear in the OpenStreetMap share of buildings, centred ($u = 0.0130 \times \text{OSM share} - 0.0074$ for `t4`); since OSM share $\approx 1 -$ ML-only share, raising it where OSM share is low means raising the gap where buildings are machine-only.

Probe	Change relative to t3	Public	Private
t4	raise predicted gap where OSM share is high, lower it where buildings are machine-only	0.08057	0.08065
t4b	the mirror image: raise it where buildings are machine-only	0.07781	0.07787
t5	t4b scaled 2.5×	**0.07655**	**0.07658**

The hidden reference moves against the objection: tracts whose buildings are machine-only have a *higher* composite gap. The quadratic identity of Section 9 gives `t5` an optimal multiplier of 1.73, i.e. the reference assigns machine-only tracts a composite gap roughly 0.03 higher than community-mapped tracts, other things equal; this was the most profitable single direction of the early phase. The leaderboard scores the composite only, so we cannot say whether the signal sits in the building or the POI component. Our final calibrated submission inherits the relationship, and it holds inside both the rural and the urban stratum, so it is not rurality in disguise:

Machine-only share of buildings	Tracts	Mean calibrated composite gap (t24)	rural tracts only	urban tracts only
< 10%	4,013	0.035	0.100	0.033
10–50%	1,893	0.054	0.082	0.049
50–90%	1,655	0.078	0.109	0.065
> 90%	1,816	0.101	0.125	0.080

If “machine-only share > 90%” were a stratum of the bias API, the API would report a disparity of **2.9×** against tracts under 10% (0.101 / 0.035), larger than the 2.36× it reports for rural versus urban. It is not a stratum, and nothing the API computes can be converted into one. More importantly, two tracts with the same composite gap of 0.10, one with 96% typeless footprints and one fully attributed, are identical to it. The content measures differ by a factor of seven between rural and urban tracts (attribute share 0.04 vs 0.30) and four (multi-source share 0.14 vs 0.53); the composite differs by 2.4×. And because a count can be raised toward a footprint reference by ingesting more detector output, the building component can be closed in these tracts without a single person ever looking at them. The scorecard’s signal and the mapping need point the same way today; the metric can nevertheless be satisfied by the wrong fix.

1.5 The POI layer shows the same provenance pattern

The places layer in rural tracts is only mildly count-deficient (40.7 POIs per 1,000 residents pooled, against 51.3 in urban tracts; tract medians 37 and 36), but its *sources* differ. In rural tracts 75% of POIs come from `meta` (Facebook business pages) against 61% elsewhere (d = 1.38); in the wildfire decile 76% versus 63% (d = 1.25). Mean confidence is lower (0.782 versus 0.816, d = −0.72) and the share of low-confidence records (< 0.5) higher (14.3% versus 11.1%, d = 0.48). Rural coverage of places rests on self-reported business pages rather than curated or community-verified records, and a count of POIs does not reveal this.

1.6 Who is affected

- **Rural residents.** 5.98 million people in the four regions live in rural tracts; 5.35 million of them in tracts where fewer than 10% of buildings carry any attribute.
- **Tribal statistical areas.** 2.64 million residents; 2.17 million in tracts under the same 10% threshold. Eastern Oklahoma's OTSAs — by tract count the Creek, Cherokee, Chickasaw, Choctaw and Kiowa–Comanche–Apache–Fort Sill Apache areas, among others — are the largest single affected population: 2.6 million people with 80% machine-only buildings.
- **Wildfire-exposed communities.** 3.33 million residents in the top exposure decile; 2.97 million under the 10% threshold. The overlap of rural and wildfire-exposed tracts holds 2.91 million people with 96.5% of footprints typeless (building-weighted).
- Socio-economic vulnerability on its own (SVI) is only weakly associated with the pattern ($d = 0.15$). This is a geography-of-contribution gap, not an income gap: it follows where volunteer mappers do and do not work.

The eight most populous tracts that are more than 95% machine-only and lie in the wildfire decile or an OTSA (from the bundle; county from the first five GEOID digits):

GEOID	County	Population	Buildings	…with a type	ML-only
48493000404	Wilson, TX	7,686	4,706	37	98%
48471790401	Walker, TX	7,277	3,365	11	99%
48091310903	Comal, TX	7,241	3,770	2	97%
48291700400	Liberty, TX	7,201	5,012	21	97%
48471790302	Walker, TX	7,092	4,202	18	96%
40009966200	Beckham, OK (Cheyenne and Arapaho OTSA)	7,078	5,524	40	97%
48099010604	Coryell, TX	6,971	3,039	3	99.7%
48339694002	Montgomery, TX	6,899	4,930	63	98%

1.7 Why it matters for disaster preparedness and climate resilience

The operational uses that the challenge motivates depend on attributes, not on footprints:

- **Evacuation planning** needs to know which footprints are dwellings and how many people a road serves. A tract whose 4,000 footprints are typeless gives the planner a count of roofs, not of households; barns, carports, pump houses and homes are indistinguishable. In the rural–wildfire overlap, 96.5% of footprints are in this state.
- **Shelter and critical-facility identification** in the building layer relies on `school`, `church`, `fire_station` and `hospital` classes, which machine-detected footprints never carry. The places layer partly compensates — rural tracts, being large, more often contain a fire-department POI (65% of rural tracts, 29% of urban ones) — but the building that houses it is untyped in the footprint layer, and the POI record is more often a self-reported page (Section 1.5).
- **Structure triage during a wildfire** (which structures are defensible and occupied) and **post-event damage assessment** both start from a typed building inventory. Where the inventory is typeless, assessors fall back to field survey, which is slowest where distances are longest.

- **Heat and climate-resilience planning** uses building type to locate manufactured housing and other heat-vulnerable stock. Tracts in the top heat-days decile carry an attribute share of 0.097 against 0.268 elsewhere.

Five places, by name (building-weighted figures from the bundle). Butte County, California, where the 2018 Camp Fire destroyed about 18,800 structures (CAL FIRE incident page), holds 105,000 Overture footprints today: 75% are machine-only and 96% carry no type. Plumas County, burned through by the 2021 Dixie Fire, is at 79% and 97%. Kerr County, Texas, where the Guadalupe River flood of July 2025 killed more than a hundred people (state and press reports), is at 67% machine-only with 84% typeless; neighbouring Gillespie County is at 94% and 99%. Cherokee County, Oklahoma, seat of the Cherokee Nation, is at 97% and 99%. Maricopa County, with 1.66 million footprints around Phoenix, is 7% machine-only. The next emergency in the first five places will be planned on a map that knows where the roofs are and not what is under them.

The consequence of the scorecard's blind spot is a policy signal that is correct in direction and silent about kind: a funder reading a rural composite gap of 0.10 is told nothing about the fact that nine in ten rural buildings are unknown, and a programme that closes the building component by ingesting more detector output would be reported as having fixed these tracts.

1.8 Recommendations

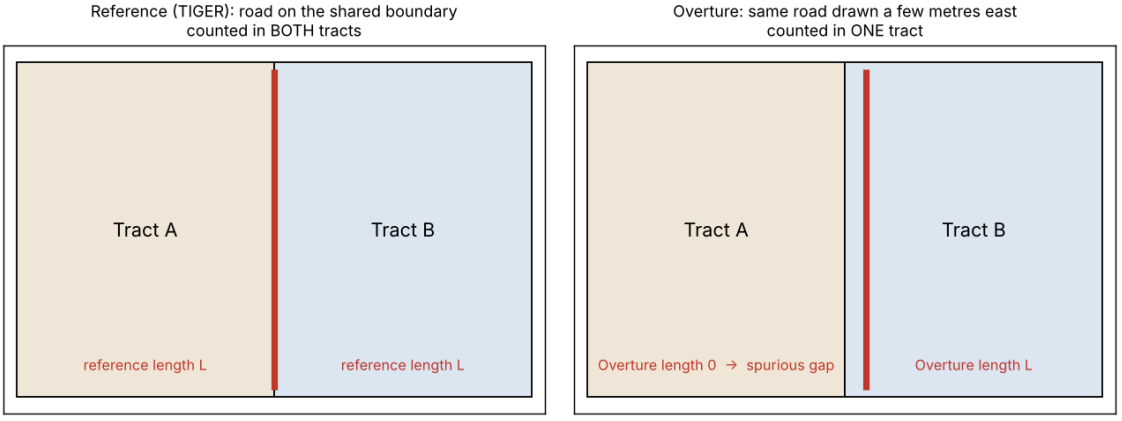
1. **Report attribute coverage alongside count coverage**: footprints relative to the reference, and the share of footprints with a type or class. The second is computable from the Overture extract alone.
2. **Report provenance with the count.** Where most candidate features originate from the reference's producer, the count comparison is not an independent measurement; publish the community-mapped and multi-source shares next to it.
3. **Add provenance strata to the bias API.** "Community-mapped share" is a stratum in its own right and, by Table 1, a more discriminating one than SVI.
4. **Direct mapping effort** at the rural–wildfire and tribal–rural overlaps named in Table 2 and Section 1.6, where 2.9 and 1.2 million people respectively live under machine-only coverage.

1.9 Limitations of Finding 1

- The leaderboard scores the composite only; the sign evidence in Section 1.4 is for the composite gap, and we cannot apportion it between the building and POI components.
- The attribute share depends on OpenStreetMap tagging practice. The multi-source share does not, and shows the same pattern (Table 1), so the conclusion does not rest on tagging conventions.
- The edition of the USFS building count in the strata table is not stated; we use it only as a second footprint inventory.
- The machine-only share is a property of the Overture release the challenge pinned (2026-08-19.0); a later release that conflates differently changes the numbers, not the mechanism.
- Groups are the scorecard's strata as provided; "rural" is `ur_class`, not a density threshold of ours. Map positions are tract bounding-box centres, not centroids.
- The external facts in Section 1.7 (fire and flood events) are context, not inputs: no number in this document depends on them.

2. Finding 2: The boundary double-count artifact

2.1 Mechanism



Schematic of the artifact.

1. Census tract boundaries frequently follow major roads: **28.9%** of our route-tagged reference length in the four regions lies within 8 m of a tract boundary (29.7% for motorways and trunks).
2. A reference road lying exactly on a shared boundary intersects the closed polygons of both tracts, so its full length counts in each.
3. Overture represents the same road a few metres off the line, so it falls inside one tract.
4. The neighbouring tract records reference length with no matching Overture length: a spurious transport gap.

Whether the organiser's pipeline counts boundary roads this way is a hypothesis about an implementation we cannot see (a strict clip of road geometry to polygons would not double count; a buffered or point-in-polygon assignment would). The leaderboard gives one test. Our model measures the reference proxy inside 8 m / 25 m buffers of the tract, which reproduces double counting; probe `c4` removed the buffers on an otherwise identical file and worsened the public score from 0.06941 to 0.06971 (private 0.06922 to 0.06967). The probe calculus puts the optimum at about one third of the way from buffered to unbuffered ($a^* = 0.35$ public, 0.29 private), and a probe with tighter buffers (`d3`) changed nothing measurable ($a^* \approx 0$). The reference therefore behaves as if boundary roads count on both sides, but not fully: consistent with double counting of some boundary segments, or with offsets smaller than our buffer. We claim the direction, not the magnitude, for the organiser's pipeline.

2.2 Size under our reconstruction, and who is affected

We computed each transport-defined tract's gap twice: with the reference measured inside the 8 m buffer (counting a boundary road in both neighbours) and strictly inside the polygon (`L0`). The difference is the artifact. Definitions: reference = core + Texas FM/RM route-tagged roads without links; numerator = all four Overture classes; defined = strict-inside reference > 100 m (n = 6,707). Under the leaderboard model's own definedness flag (core routes within buffers > 100 m, n = 6,269) the numbers are close: mean +0.061, rural 0.095 versus 0.050 (1.89×). Full table: `results/bias_artifacts_table.csv`; all other numbers: `results/bias_artifacts_numbers.json`.

Group	Artifact, in group	rest	Ratio	Welch t	p	Cohen's d
Rural	0.099	0.055	**1.81×**	11.2	$< 10^{-10}$	+0.32
Tribal statistical area	0.084	0.064	**1.32×**	3.2	0.001	+0.15
High SVI minority (≥ 0.75)	0.055	0.072	0.76×	−4.9	$8 \cdot 10^{-7}$	−0.13
High SVI overall (≥ 0.75)	0.064	0.067	0.95×	−1.0	0.31	−0.03

- Across defined tracts the artifact adds a mean of **+0.066** to the transport component, and more than 0.05 in **27%** of them (19% of all tracts).
- **Why it concentrates in rural tracts.** Not exposure: rural tracts have *less* boundary road (34% of their reference length lies within 8 m of a boundary, against 58% in urban tracts; the buffer adds a median 11% to their reference length, against 30% in urban tracts). Sensitivity: in urban tracts the four-class Overture length exceeds the route-tagged reference by a median factor of 2.4, so the gap is clipped at zero for 96% of them and extra reference length changes nothing; rural tracts sit at a median ratio of 1.0, where 43% have a positive gap and every double-counted metre becomes gap. The artifact bites where the metric is live, and the metric is live in rural tracts.
- **Propagated to the composite** (weight 1/3 in defined tracts), removing the artifact from our final submission lowers the rural/urban ratio of mean coverage gap from 2.36× to 2.20× and removes 29% of the rural–urban difference (30% under `t19`). For tribal areas the absolute difference falls 13%, but the ratio — which is what the API reports — *rises* from 2.78× to 3.03×, because the artifact removed outside tribal areas is a larger fraction of a smaller base. The artifact distorts the rural disparity downward when removed and the tribal ratio upward; it is not a uniform inflation.

2.3 Recommendation

Assign boundary-coincident reference roads to one tract, or split them 50/50, before computing ratios; and report the transport disparity with and without the correction so that users can separate mapping gaps from measurement effects. Because the artifact's size depends on how far each tract sits from the clipped-at-zero region, report it per stratum rather than as one number.

3. Finding 3: Ramps and link roads

Overture carries ramps and connector roads as `subclass = link` within the counted classes (by length: 73% `motorway`, 8% `trunk`, 12% `secondary`, 7% `primary`); TIGER codes ramps S1630, outside the S1100/S1200 reference. If the organiser's numerator counts links — the methodology defines it by class, so it does — ramp length is credited against a reference that excludes ramps, which hides gaps in interchange-heavy tracts.

- Under our reconstruction the effect is small on average (+0.004 on the transport component), **3.7× larger in rural tracts** (0.0094 versus 0.0026, t = 10.6), where a single interchange is a large share of named-highway length, and **0.2× in tribal areas** (0.0010

versus 0.0046, t = −9.2) and 0.5× in high-minority tracts, which have denser non-link road networks.
- Links are a larger share of all four-class Overture length in high-SVI-minority tracts (9.2% versus 6.9%, tract means) and a smaller share in rural tracts (3.4% versus 8.7%).
- **This finding was not tested on the leaderboard.** A probe with links removed from the numerator (`c3`) was prepared but never scored. The QGIS inspection in the Appendix examined the 200 longest `motorway`/`trunk` segments in Maricopa that carry no route tag — segments a class-based numerator counts and a route-based reference does not — and found them to be interchange ramps, collector–distributor lanes, frontage/service connectors and untagged second carriageways. That documents where the two definitions diverge on the ground; it does not measure the organiser's pipeline.

Recommendation: exclude `subclass = link` from the numerator so that it matches TIGER's exclusion of S1630, and report the tract-level link share so that readers can see where the exclusion matters.

4. Reproducibility of Part I

Script	Inputs	Writes
`code/quality_gap.py <bundle> <out> [t24.csv]`	bundle; optionally the final submission	`quality_gap_table.csv` (Table 1 and Section 1.5), `quality_gap_regions.csv` (Table 2), `quality_gap_numbers.json` (every other number in Section 1, including the county figures and the t24 bin table), the three figures of Section 1
`code/bias_artifacts.py rb3.parquet <bundle> <out> [t24.csv t14.csv c4.csv roadfeat2.parquet]`	shipped geometry-step output, bundle, optional submissions	`bias_artifacts_table.csv` (Section 2.2 table), `bias_artifacts_numbers.json` (28.9%, exposure/sensitivity figures, link shares, composite propagation, c4 a*, model-flag variant)

Both run in a few minutes with pandas, numpy and matplotlib (`requirements.txt`). No external data, no network access, no stochastic step. The 200 QGIS screenshots are available on request; four are in the Appendix.

PART II — METHODOLOGY DOCUMENTATION

5. Data and rules compliance

- **Inputs**: the bundle only — `*-tracts`, `*-strata` (227 columns), `*-acs`, the Overture roads, buildings and places extracts as distributed — plus aggregates we computed from them (per-tract road lengths with boundary buffers, building counts by source, POI counts by category).
- **Not used**: TIGER/Line, Microsoft Building Footprints, HIFLD, County Business Patterns, or any copy of them, for scoring or for the discovery analysis.

- **Scope**: the submission template has 9,794 rows: 9,379 tracts in the four scored regions (the bundle's tract table has seven more, water-only "Census Tract 9900" polygons in Texas with no population and no buildings, which are not in the template) and 415 tracts in Eastern Washington. Part I uses the bundle's 9,386 tracts; Part II's edge-case counts (Section 11) are over the 9,379 scored ones.
- **Tools**: Python 3, pandas, numpy, DuckDB (spatial extension). Geometry ran on Kaggle CPU kernels in EPSG:5070 (CONUS Albers, equal-area, metres).
- **Determinism**: no step is stochastic; every intermediate file reproduces exactly.

6. Structure of the reference score

From the published methodology, for tract *i* with building gap B, POI gap P and transport gap T:

$$y_i = \frac{B_i + P_i + m_i T_i}{2 + m_i}, \qquad m_i = \mathbb{1}\left[\text{reference highway length in tract} > 0\right].$$

We treat B and P as defined in every tract. The methodology reports the share of tracts with at least one undefined component as 21% (OK), 37% (CA), 28% (TX) and 55% (AZ); our transport-undefined shares under the model's definition are 22%, 39%, 29% and 55%. The two agree within two points in every region, which leaves no room for a second undefined component of any size.

7. Transport component reconstruction

Numerator. Length of `motorway`, `trunk`, `primary` and `secondary` segments inside the tract (`ST_Length(ST_Intersection(road, tract))`, column `L0`), all subclasses.

Reference proxy. TIGER S1100/S1200 are numbered Interstate, US and State routes. We flag an Overture segment as `core` when any of its `routes[].network` values matches (exactly as in `code/routes4.py`):

```
^US:I(:.*)?$            (excluding networks containing "Future")
^US:US(:(Business|Alternate|Spur|Bypass|Truck))?$
^US:(AZ|CA|OK|TX)(:(Loop|Spur|Business|Beltway|Park))?$
^US:(OK:Turnpike|TX:Toll|TX:NTTA|TX:Harris:HCTRA)$
```

and as `fm` when it carries a Texas FM/RM network (`US:TX:FM`, `US:TX:RM`), computed in the geometry step. Ramps (`subclass = link`) are excluded from the reference proxy, because TIGER codes them S1630.

Boundary buffers. The proxy is measured inside the tract polygon buffered by 8 m (all classes) and 25 m (`motorway`, `trunk`): columns `L8`, `L25`. `Lbd8` is the length within 8 m of the boundary itself, used in Part I. The geometry step (per segment × tract, in EPSG:5070) is shipped as its output `data/rb3.parquet`; the DuckDB query is reconstructed in `code/geometry_step.sql`, since the Kaggle kernel itself was not preserved.

Gap and definedness. T = clip(1 − numerator / reference, 0, 1); m = 1 when the core-route reference within the buffers exceeds 100 m. The threshold is insensitive: 50 m or 250 m would change 13 and 39 tracts. Texas FM/RM roads enter the reference scale as separately calibrated directions (`a3`, `u2`), not through the definedness flag. The reference scale itself was calibrated per region on the leaderboard (`e1`, `e2`, `f1`–`f3`).

Overture tagging gap. Rescuing untagged mainline carriageways identified in the QGIS inspection changes only 94 tracts (about 1,000 km against 52,000 km of tagged core).

8. Building and POI proxies, and the final predictor

- **Building directions**: Overture counts by source (OpenStreetMap share, Microsoft share); counts without small structures (< 40 m²) against `usfs_BuildingCount_sum`; area per structure; multi-source share.
- **POI directions**: places per capita and per housing unit; source shares and confidence; facility categories (fire, EMS, schools; hospitals excluded, as the methodology states); business-type places relative to housing units; county-level POI per capita.
- **Neighbourhood**: 8-nearest-tract means of the prediction within each region (tract bounding-box centres, longitude scaled by cos latitude), applied to transport-undefined tracts.
- **Region and definedness constants, region stretch, rural indicators**: the late orthogonal probes (`rc*`, `ru*`, `st*`).

The final predictor is a linear combination of scored submissions: **t24 = t23 + Σₖ aₖ (sₖ − t23)**, clipped to [0, 1], over 76 scored vectors s_k. The coefficients a_k, with each direction's noise estimate σ_k and the quantities q_k and g_k of Section 9, are in `results/t24_coefficients.csv`; the scored vectors are shipped in `data/scored_vectors.parquet` with their public and private scores in `data/scores.csv`; `code/make_t24.py` rebuilds t24 from them (maximum absolute difference from the submitted file $1.8 \cdot 10^{-5}$, which comes from scores rounded to six digits for the earliest files at the time).

9. Leaderboard-probe calculus

Let *s* be a scored submission with public MSE M_0 and *u* a perturbation vector over the publicly scored tracts. Then

$$\mathrm{MSE}\,(\,s + au\,) = M_0 + 2a\,c + a^2 q, \qquad q = \bar{u^2}, \quad c = u\,(\,\bar{s} - y\,),$$

with bars denoting means over the scored tracts. *q* is computed locally; one scored probe at a = 1 gives $c = (M_1 - M_0 - q)/2$ and the optimal step $a^* = -c/q$. For many directions we solve $Qa = -c$ with the local Gram matrix $Q = (\langle u_j, u_k \rangle)$; information from earlier probes carries over to a new base exactly through c_new = c_old + ⟨u, Δbase⟩ (mean inner product).

9.1 Global re-fit with a noise model

Every scored file is a vector with one known inner product against the hidden residual. We re-fit the best combination of all scored vectors at once:

$$a^* = -\left(G + \lambda\,\mathrm{diag}\left(\sigma_k^2\right)\right)^{-1} g,$$

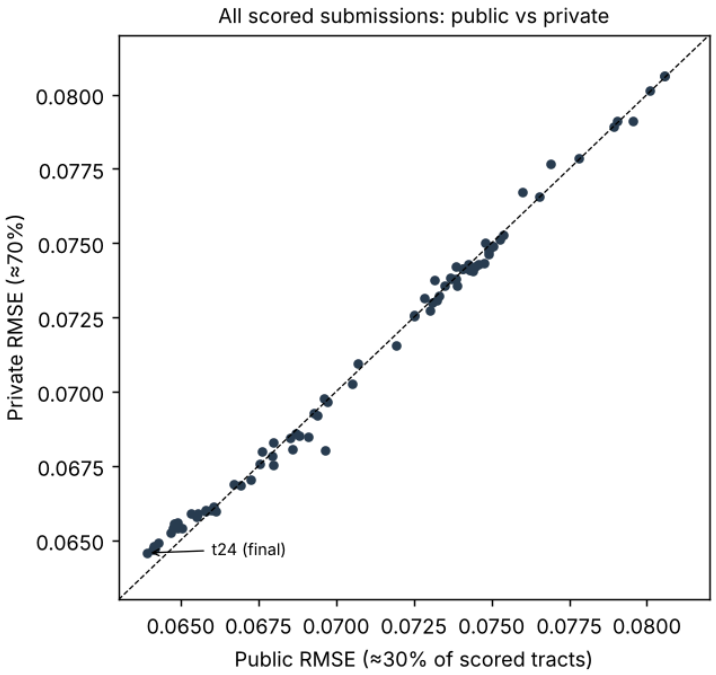
where D is the matrix whose columns are $d_k = s_k - s_{base}$, $G = D^{\mathsf{T}}D / n$ is its Gram matrix over the scored tracts, $g_k = (M_k - M_{base} - q_k)/2$ is the inner product of d_k with the hidden residual implied by the quadratic identity, σ_k is estimated by leave-one-out prediction of each scored submission from all the others (a vector whose score the others cannot predict is treated as noisier; $\sigma_k = 2 \cdot 0.066 \cdot \max(\text{LOO error}, 2 \cdot 10^{-5})$), and $\lambda = 3 \cdot 10^5$ was chosen by simulation so that the *expected realised* gain under noisy g is maximised rather than the in-sample gain. The late orthogonal probes (`rc*`, `ru*`, `st*`) were given their measured, smaller noise ($3 \cdot 10^{-5}$).

9.2 Quality control that changed results

- **Leave-one-out consistency revealed two mislabeled batches** (scores attached to the wrong files: b1/b2/b4, and a1/a2). With the corrected mapping the joint prediction error for `t14` fell from 0.00102 to 0.00021, and the re-fit `t18` was predicted at 0.06601 and scored 0.06599.
- **Prediction accuracy of the calculus on the last four re-fits:** `t21` predicted 0.06486 → scored 0.06490; `t22` 0.06464 → 0.06476; `t23` 0.06408 → 0.06415; `t24` 0.06390 → 0.06391.
- **Eastern Washington.** A probe that changed only Eastern WA tracts (`f4`, 128 tracts, RMS change 0.105) left the public score unchanged to nine digits, so all public inner products are computed on the four regions. After the close its *private* score differed from its base (0.06831 versus 0.06755): **Eastern WA is excluded from the public board but included in the private one.** Our WA predictions were never calibrated; the re-fits moved them by an RMS of 0.012 between `t18` and `t24`. We state this because it bounds what Section 9.3 can conclude.
- **Public/private.** The public board is a fixed subset of the four regions; the private board adds the remaining tracts and Eastern WA. We shrink directions in proportion to their estimated noise; Section 9.3 audits how well that worked.

9.3 Where the calculus overfit the public split

With the private scores visible after the close, the method can be audited. Across all 85 scored submissions the public and private RMSE correlate at 0.9986 (median |difference| 0.00023). The structural improvements (`t3` → `t18`) transferred almost one-for-one: `t18` scored 0.06599 public and 0.06600 private. The probe-driven re-fits after it did not transfer fully: the private–public difference grows from +0.00001 (`t18`) to +0.00019 (`t19`), +0.00025 (`t20`), +0.00050 (`t21`) and +0.00066 (`t22`–`t24`). Of the 0.0021 public gain from `t18` to `t24`, about 0.0014 appeared on the private board. Two effects are confounded in the remaining 0.0007: fitting to the public sample, and the uncalibrated Eastern WA predictions that the private board includes; the `t16`/`f4` pair shows WA alone can move a private score by 0.0008. The noise model anticipated the direction of the first effect but not its size. `t24` is still the best private submission of the 85 (0.06457), so the overfit reduced the gain rather than reversing it.



Public against private RMSE for all 85 scored submissions; the dashed line is equality.

10. Results (RMSE on `coverage_gap_score`)

Submission	Content	Public	Private
`t3`	transport reconstruction with 8/25 m boundary buffers	0.07904	0.07911
`t5`	+ OSM-share direction (Section 1.4)	0.07655	0.07658
`t7–t9b`	+ source-mix and per-capita POI/building directions	0.07430	0.07411
`t10–t13`	+ structural building (USFS count), TX FM roads, denominator weighting	0.07251	0.07257
`t14`	+ primary/secondary routed roads in reference (largest single direction)	0.06941	0.06922
`t15–t17`	+ buffer and reference-scale calibration by region	0.06792	0.06782
`t18`	global re-fit, corrected labels	0.06599	0.06600
`t19`	+ county/kNN/tail/facility probes	0.06581	0.06601
`t20`	noise-weighted re-fit over all scored vectors	0.06552	0.06577
`t21–t22`	re-fits including kNN probes	0.06476	0.06542
`t23`	+ region × definedness constant probes	0.06415	0.06481
`t24 (pwM1ZRTe)`	+ rural and stretch probes	**0.06391**	**0.06457**

Directions that did not help, reported for completeness: Texas FM roads added to the reference alone; rescaling building/POI corrections by 1/(2+m) (the opposite sign was optimal); population-based facility-absence indicators (optimal weight ≈ 0); county-level POI-per-capita gap (negative weight); kNN smoothing applied to transport-defined tracts (optimal weight ≈ 0, probe `k6`); scaling Eastern WA (`f4`, no public effect, worse private).

Leaderboard caveat. The two files marked as final on the submissions page were set on 11 October and never updated, so the closing leaderboard shows dodo25 at 0.0795 / 0.0791. The method's final file is `pwM1ZRTe` (0.06391 public, 0.06457 private), which the judges can verify on the submissions page. We state this plainly rather than let the writeup disagree with the board.

11. Edge cases and alternative weightings

The prize text asks how zero-population, water-dominated and data-sparse tracts were handled. The table gives their counts among the 9,379 scored tracts of the four regions (the seven water-only tracts absent from the template are excluded) and the mean prediction of `t24` for each (overall mean 0.059).

Edge case	Definition	Scored tracts	Mean prediction	Handling
Zero population	`pop_total = 0`	26	0.145	Per-capita ratios are undefined. In code a zero or missing denominator is clipped to 1 and an undefined ratio takes its neutral value (gap contribution 0), so these tracts are predicted from the transport reconstruction, the building count against the USFS count, and the 8-NN neighbourhood mean. They receive the highest mean prediction; we regard this as a known weakness, since the reference itself is least meaningful where nobody lives.
Near-zero population	`pop_total < 100`	65	0.093	Same clipping. Small denominators are the main reason these predictions sit above average; accepted because the tracts are few and contribute little to RMSE.
Zero housing units	ACS `housing_units = 0`	80	0.098	Housing-unit ratios take their neutral value; building proxies use the USFS count instead.
Water-dominated	land area < 50% of tract area (`ALAND` / `area_m2_calc`)	43	0.114	A land-area density direction (g2) and a water-share direction (g4) were

Edge case	Definition	Scored tracts	Mean prediction	Handling
				prepared but never scored, so these tracts receive no special treatment beyond the neighbourhood mean. Listed as an open item. The seven water-only tracts are not in the template.
No Overture buildings	`n = 0`	2	0.097	Building ratios undefined; prediction from transport and neighbourhood.
Data-sparse	< 20 buildings / no POIs	5 / 2	0.105 / 0.097	Ratios computed with clipped denominators; the neighbourhood term supplies most of the signal.
USFS count = 0	`usfs_BuildingCount_sum = 0`	2	0.097	Building direction contributes zero.
Transport undefined	core-route reference within buffers < 100 m	3,110	0.023	Score structure (B + P)/2; 8-NN smoothing within region with optimal weight 0.68 on this subset (probe k4).
Bounds	—	all	—	Predictions clipped to [0, 1]. Early files used a 0.01 floor (1,450 tracts). Probe d4 set those tracts to 0 and received an optimal weight of −0.68: the floored tracts were under-predicted, not over-predicted, and later re-fits moved them up. 124 tracts of the final file sit at

Edge case	Definition	Scored tracts	Mean prediction	Handling
				exactly 0 through clipping of the combined fit.
Eastern WA	415 tracts	—	—	Predicted with the same pipeline but never calibrated: excluded from the public board, present on the private board (Section 9.2).

11.1 Assumptions, stated

1. Only `coverage_gap_score` is scored: probes that changed only the component columns left the score unchanged.
2. The score has the structure of Section 6, with building and POI components defined in essentially every tract (argument in Section 6).
3. Transport is undefined where the core-route reference is below 100 m; insensitive to the threshold (Section 7).
4. The public board is a fixed subset of the four regions' tracts, so inner products estimated on it are unbiased for those regions up to sampling noise; the private board adds the remaining tracts and Eastern WA (Section 9.2). Section 9.3 reports how far the transfer held.
5. Predictions are bounded in [0, 1], as the reference is a ratio-based score.
6. The reference is deterministic, so the quadratic identity of Section 9 holds exactly on the public sample.

Alternative weightings tested. Weighting building/POI directions by each tract's actual 1/(2+m) share versus a uniform weight: uniform was better, which suggests the organiser's B and P vary more in transport-defined tracts. Region-specific reference scales (`e1`, `e2`, `f1`–`f3`) were kept; a county-level POI direction was rejected.

12. Code map and how to reproduce

Repository: `github.com/tdogukan470-jpg/zindi-bias-bounty-dodo25` (public). Python 3.10+, `pip install -r requirements.txt`.

Path	Purpose
`data/rb3.parquet`	Output of the geometry step: per (tract, class, route flags, subclass) lengths L0/L8/L25/Lbd8 in EPSG:5070
`data/roadfeat2.parquet`	Per-tract road features derived from `rb3` (`coreB` is the model's definedness reference)
`data/scored_vectors.parquet`, `data/scores.csv`	All 78 scored submission vectors with Zindi ids and public/private scores (85 rows; 7 early or duplicate uploads have no named vector)
`code/geometry_step.sql`	Reconstructed DuckDB query of the geometry step (the Kaggle kernel was not preserved; its output is `rb3.parquet`)
`code/routes4.py`	Route-tag regex over `routes[].network` (historical; reads the per-segment table built on Kaggle)
`code/feat.py`, `code/cands.py`	Historical feature and direction scripts (read intermediate files from the Kaggle step)
`code/probe_solve.py`	Quadratic probe solver, ridge joint fit
`code/make_t24.py`	Rebuilds the final submission from the scored vectors and scores; writes `results/t24_coefficients.csv`
`code/quality_gap.py`	Part I, Finding 1: tables, numbers and figures
`code/bias_artifacts.py`	Part I, Findings 2–3: tables and numbers
`code/leaderboard_audit.py`	Section 9.3: `results/all_submissions_public_private.csv`, `figs/fig_public_private.png`
`results/`	`quality_gap_table.csv`, `quality_gap_regions.csv`, `quality_gap_numbers.json`, `bias_artifacts_table.csv`, `bias_artifacts_numbers.json`, `t24_coefficients.csv`, `all_submissions_public_private.csv`
`submissions/`	Scored files `t18`–`t24`

Order: `geometry_step.sql` (Kaggle) → `rb3.parquet` → model directions → scored submissions → `make_t24.py`. Part I scripts need only the bundle, `rb3.parquet`, and the shipped submissions.

References

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- Jaffe, Scott, Callahan, Dillon, Karau, Lazarz (2024), *Wildfire Risk to Communities: Spatial datasets of wildfire risk for populated areas in the United States*, 2nd ed., USFS Research Data Archive, doi:10.2737/RDS-2020-0060-2 (metadata retrieved 1 Nov 2026).

- CAL FIRE, Camp Fire incident page (structures destroyed). https://www.fire.ca.gov/incidents/2018/11/8/camp-fire (retrieved 1 Nov 2026); Dixie Fire incident page, same site.
- Texas flood of July 2025: Office of the Governor and press reports put the Kerr County toll above 100 (e.g. Texas Public Radio, 8 Jul 2025, https://www.tpr.org/news/2025-07-08/more-than-100-people-died-in-texas-historic-floods-here-are-some-of-their-stories; retrieved 1 Nov 2026).

Appendix — QGIS field inspection (Maricopa)

The 200 longest `motorway`/`trunk` segments without a route tag were each rendered over the OpenStreetMap basemap with the Overture road layer styled by class (basemap © OpenStreetMap contributors). Four representative cases follow.

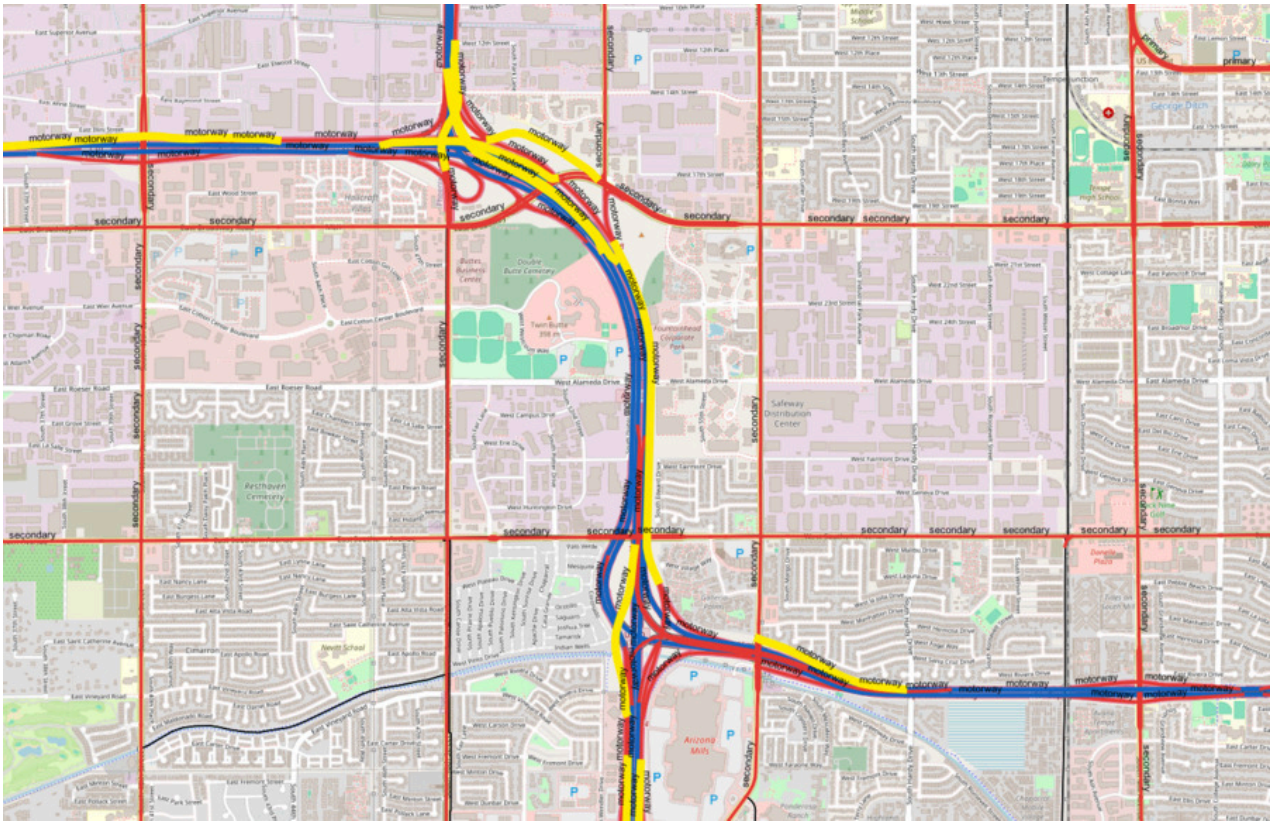


Fig. A1 — Freeway-to-freeway interchange (segment 6348, 1.8 km). Ramps and collector lanes branch from the mainline; Overture classes them `motorway`, TIGER codes them S1630.

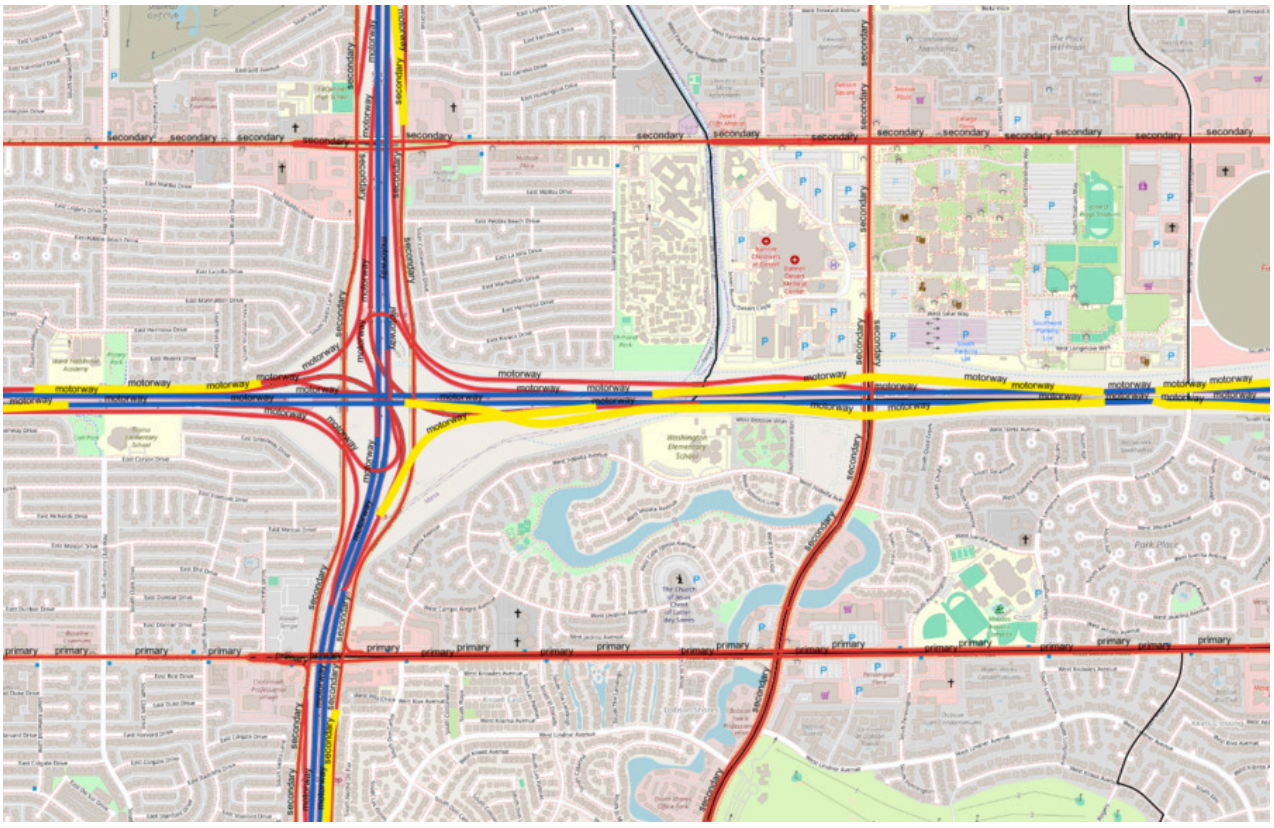


Fig. A2 — Interchange on an arterial grid (segment 9798, 1.4 km). Ramp loops and frontage/service connectors run parallel to the mainline.

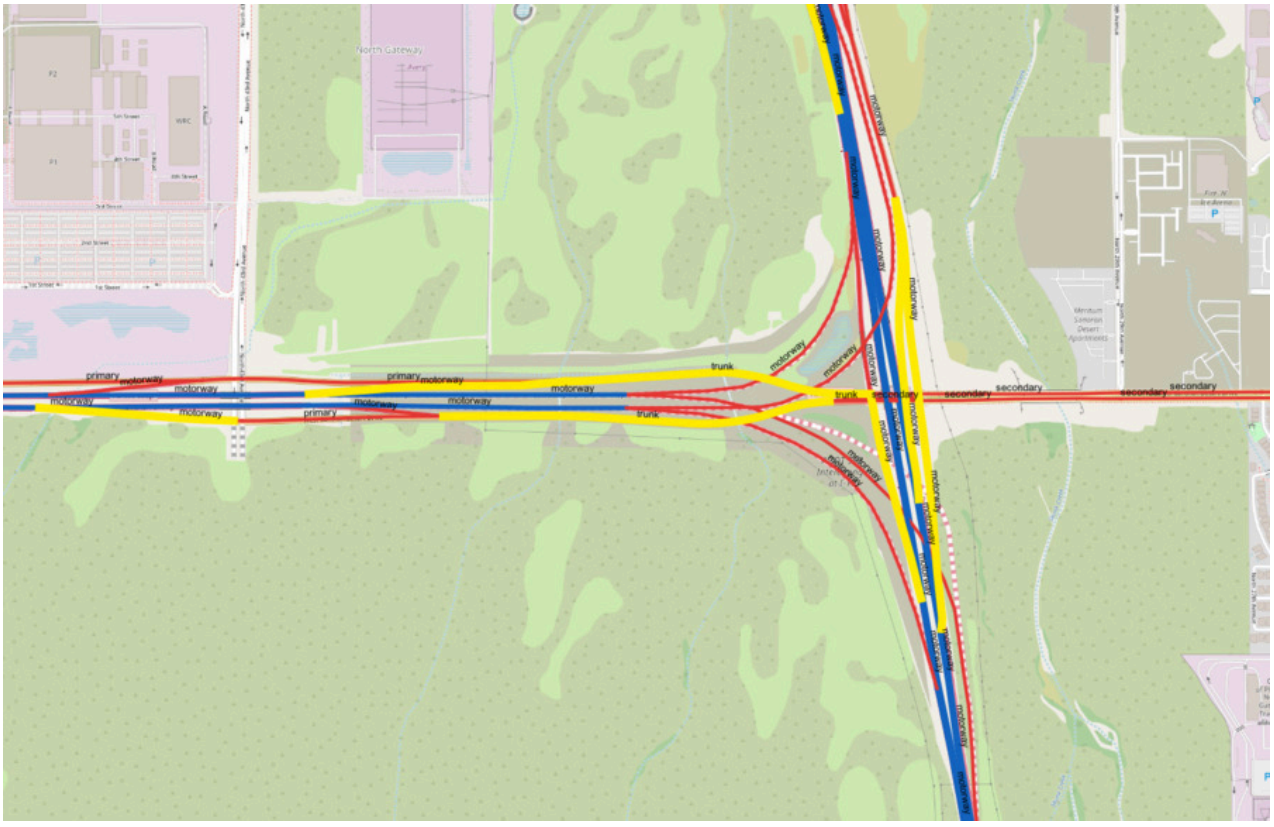


Fig. A3 — Rural trunk junction (segment 30157, 1.2 km). Link roads make up a large share of named-highway length in a sparse tract.

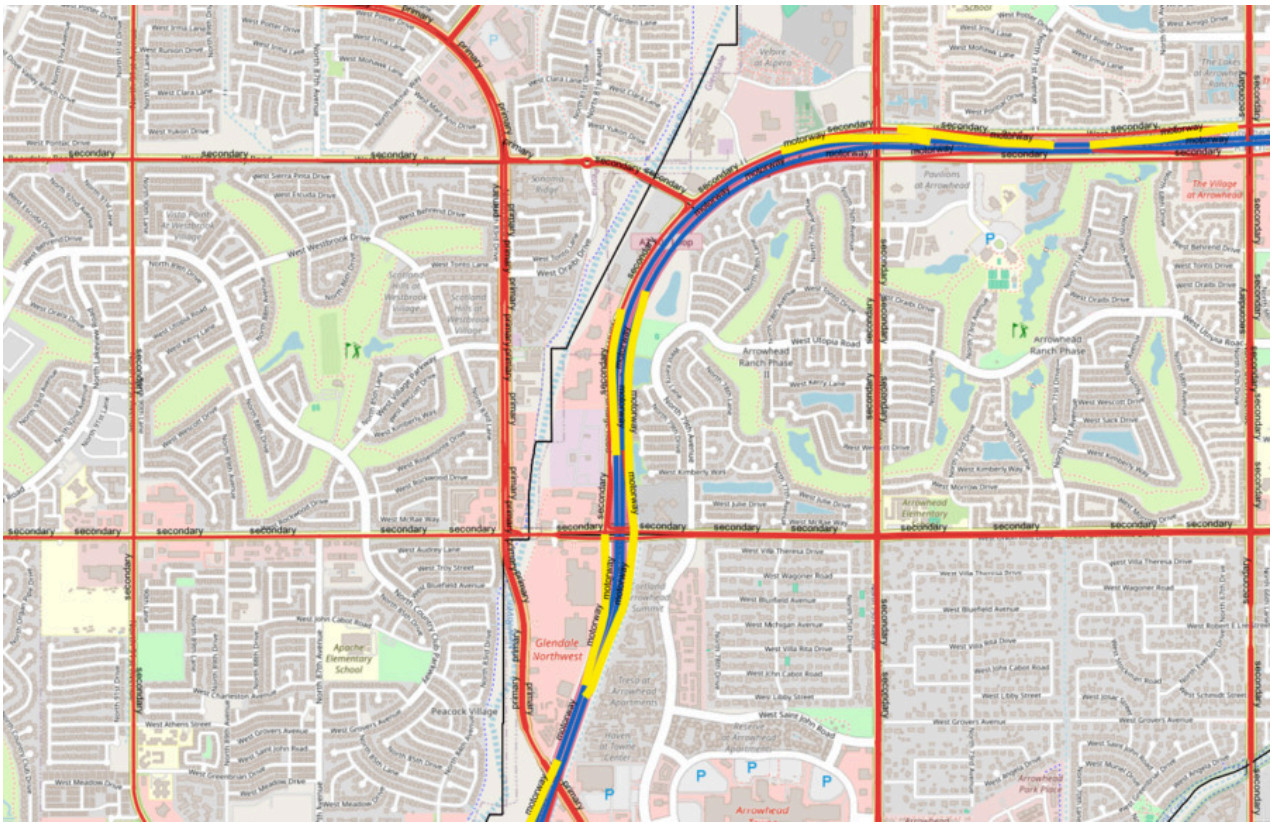


Fig. A4 — Curving divided freeway (segment 28771, 1.0 km). The second carriageway carries no route tag, so a route-based reference proxy misses it.